

Feature-based Alignment

Chapter 6

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- ★ 2D and 3D feature-based alignment
- ★ Pose estimation
- ★ Geometric intrinsic calibration

Introduction

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- ★ At the topic of geometric image registration
 - The computation of 2D and 3D transformations that map features in one image to another.
- ★ Special case of this problem is pose estimation
 - Determining a camera's position relative to a known 3D object or scene.
- ★ Camera's intrinsic calibration
 - Internal parameters such as focal length and radial distortion.

Introduction

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Matching Geometric Models via Alignment

- ★ Alignment is the most common paradigm for matching 3D models to either 2D or 3D data.
- ★ The steps are:
 1. *Hypothesize a correspondence* between a set of model points and a set of data points.
 2. From the correspondence *compute a transformation* from model to data.
 3. *Apply the transformation* to the model features to produce transformed features.
 4. *Compare* the transformed model features to the image features to verify or disprove the hypothesis.

2D and 3D Feature-based Alignment

Feature-based Alignment

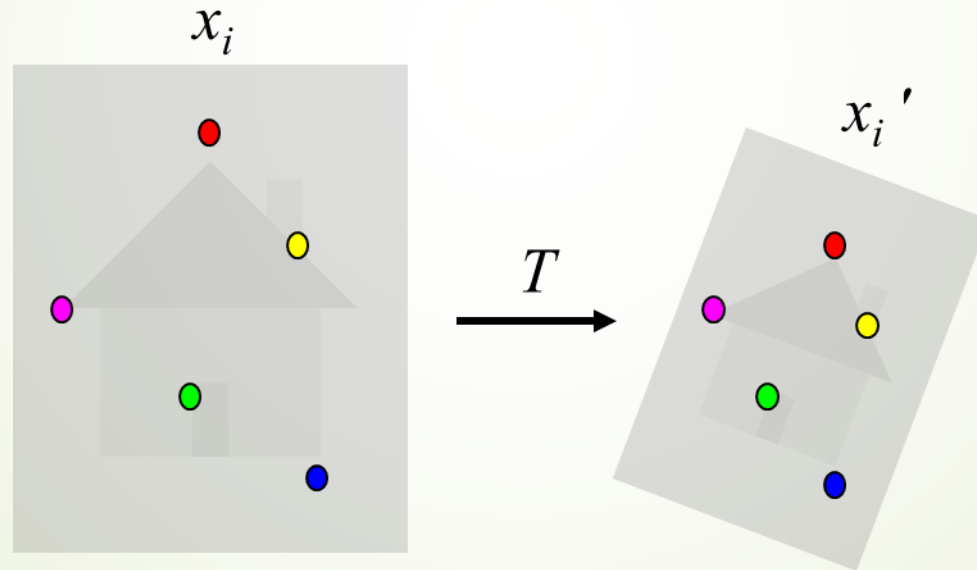
- ★ Match extracted features across different images
- ★ Verify the geometrically consistent of matching features
- ★ Applications:
 - Image stitching
 - Augmented reality
 - ...

2D and 3D Feature-based Alignment

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Alignment Problem

- ★ In alignment, the parameters of some transformation will be fit according to a set of matching feature pairs (“correspondences”).



2D and 3D Feature-based Alignment

Image Alignment

★ Two broad approaches:

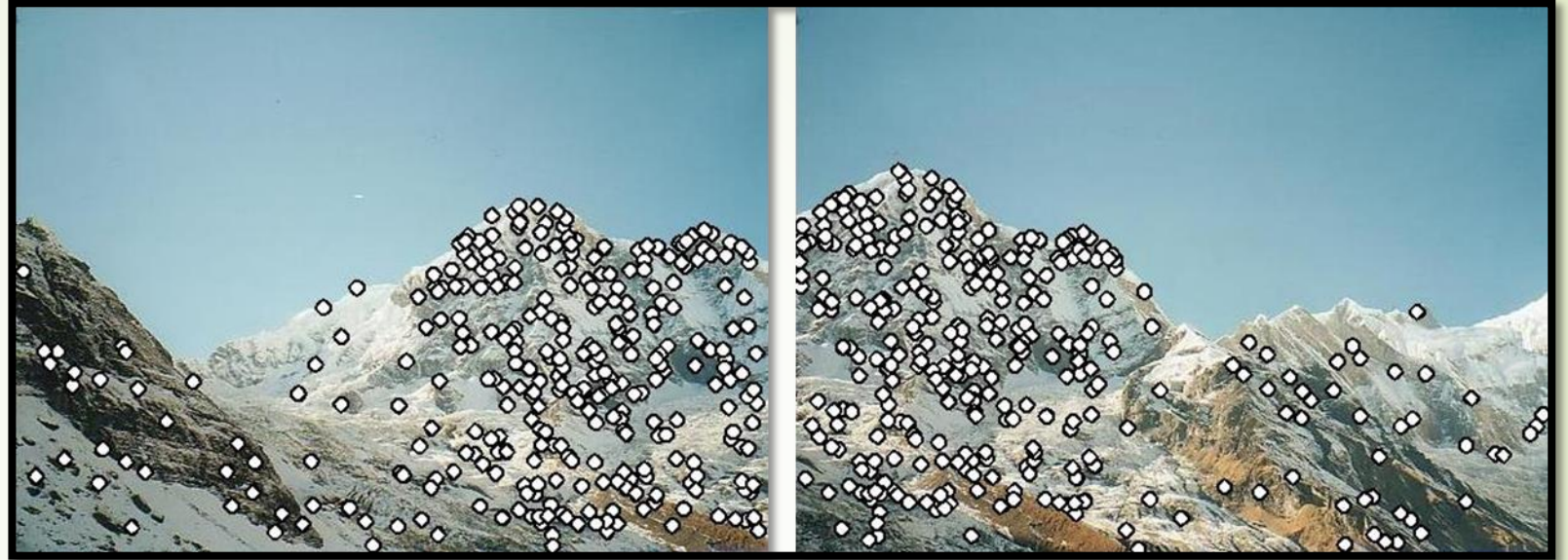
- Direct (pixel-based) alignment
 - Search for alignment where most pixels agree
- Feature-based alignment
 - Search for alignment where extracted features agree
 - Can be verified using pixel-based alignment



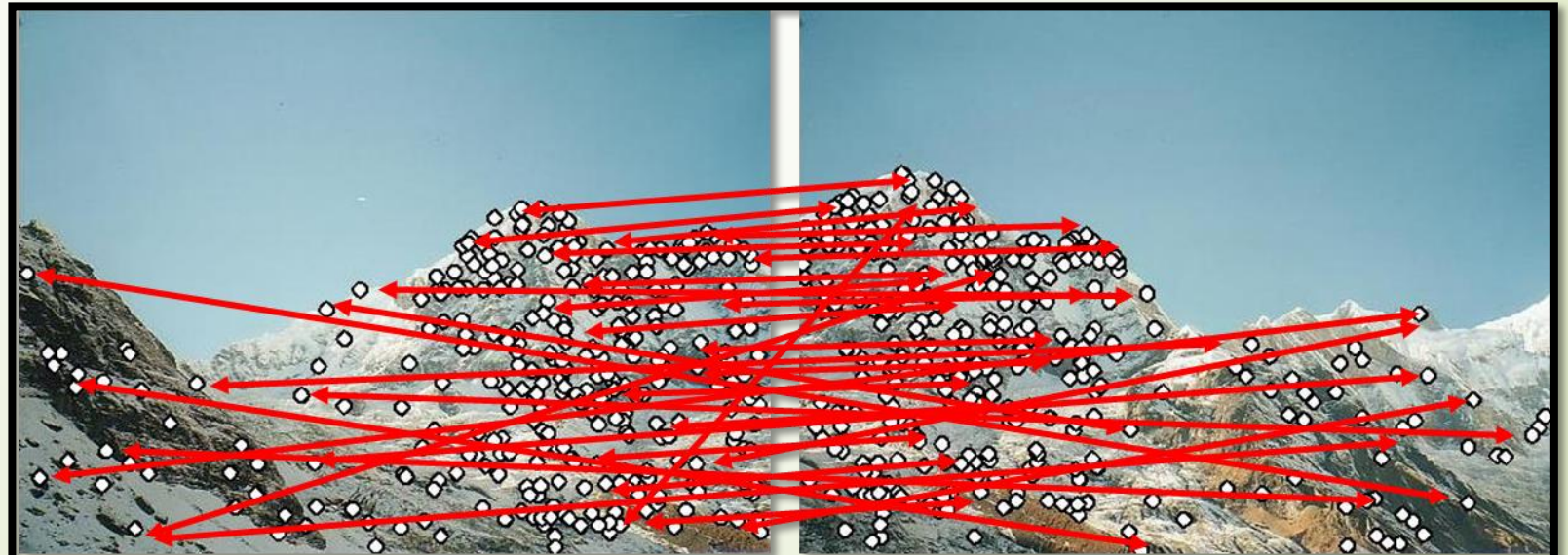
2D and 3D Feature-based Alignment

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★ Extract Features



★ Compute the matches



2D and 3D Feature-based Alignment

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Final Result

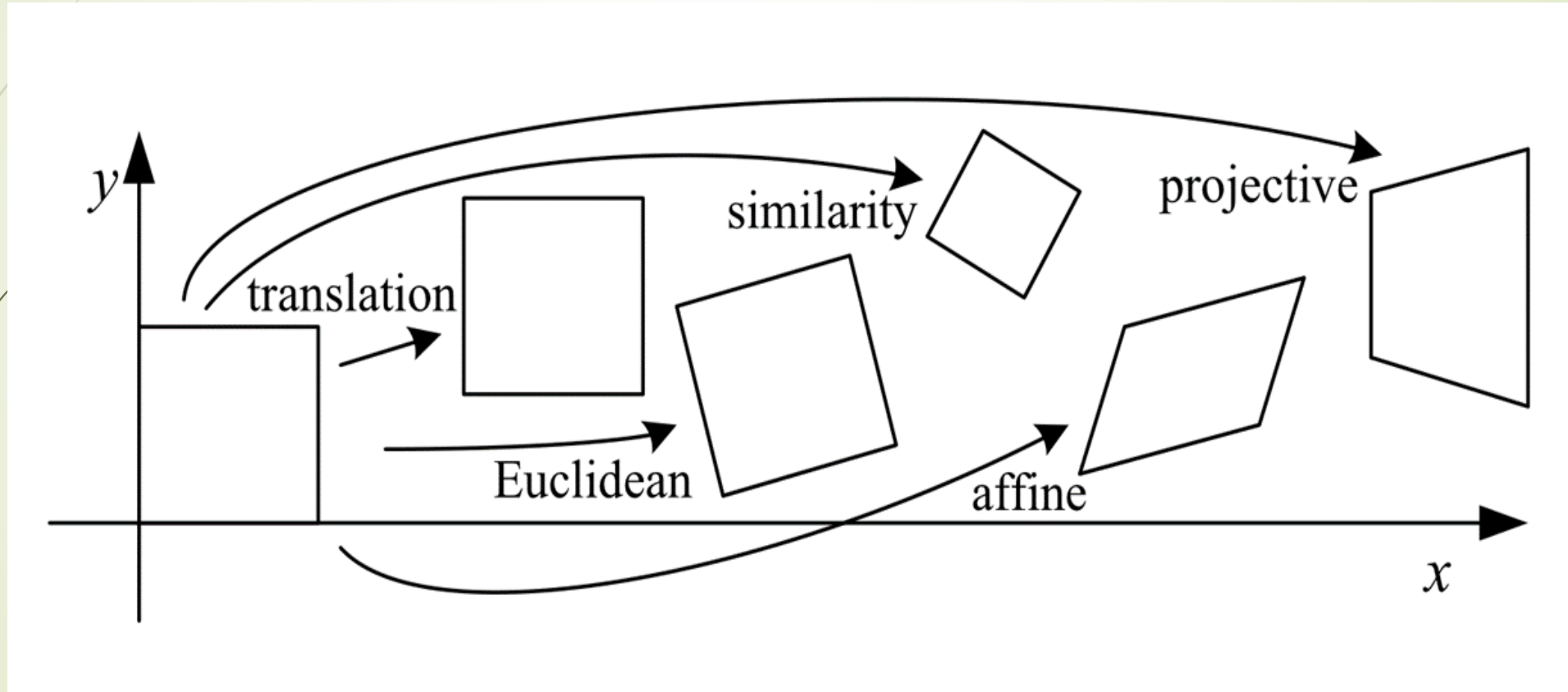


Loop:

- ★ Hypothesize transformation T
- ★ Verify transformation (search for other matches consistent with T)

2D and 3D Feature-based Alignment

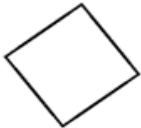
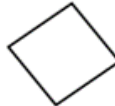
2D Planar Transformations



Basic set of 2D planar transformations

2D and 3D Feature-based Alignment

2D Planar Transformations

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} \mathbf{I} & & \mathbf{t} \end{bmatrix}_{2 \times 3}$	2	orientation	
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & & \mathbf{t} \end{bmatrix}_{2 \times 3}$	3	lengths	
similarity	$\begin{bmatrix} s\mathbf{R} & & \mathbf{t} \end{bmatrix}_{2 \times 3}$	4	angles	
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}_{2 \times 3}$	6	parallelism	
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}_{3 \times 3}$	8	straight lines	

2D and 3D Feature-based Alignment

2D Alignment Using Least Squares

- ★ Given a set of matched feature points $\{(x_i, x'_i)\}$
- ★ A planar parametric transformation:

$$x' = f(x; p)$$

where, p are the parameters of the function f

How to estimate the motion parameters p ?

- ★ Residual:

$$r_i = x'_i - f(x_i; p) = \hat{x}_i - \tilde{x}_i$$

where, $\hat{x}_i$: the measured location

$\tilde{x}_i$: the predicted location

2D and 3D Feature-based Alignment

2D Alignment Using Least Squares

★ Least squares:

➤ Minimize the sum of squared residuals

$$E_{LS} = \sum_i \|r_i\|^2 = \sum_i \|f(x_i; p) - x'_i\|^2$$

★ Many of the motion models have a linear relationship:

$$\Delta x = x' - x = J(x)p$$

★ $J = \partial f / \partial p$: The jacobian of the transformation f .

2D and 3D Feature-based Alignment

Iterative Algorithms

- ★ Most problems do not have a simple linear relationship between the measurements and the unknowns.
- ★ The resulting problem is called *non-linear least squares* and *non-linear regression*.
- ★ Iteratively find an update Δp to the current parameter estimate p by minimizing:

$$\begin{aligned} E_{\text{NLS}}(\Delta p) &= \sum_i \|f(x_i; p + \Delta p) - x'_i\|^2 \\ &\approx \sum_i \|J(x_i; p)\Delta p - r_i\|^2 \\ &= \Delta p^T \left[\sum_i J^T J \right] \Delta p - 2\Delta p^T \left[\sum_i J^T r_i \right] + \sum_i \|r_i\|^2 \\ &= \Delta p^T A \Delta p - 2\Delta p^T b + c, \end{aligned}$$

2D and 3D Feature-based Alignment

Robust Least Squares

- ★ Ordinary least squares estimators are sensitive to the presence of observations that lie outside the norm for the regression model of interest.
- ★ Robust least square refers to a variety of regression methods designed to be robust, or less sensitive, to outliers.
- ★ Three different methods for robust least squares:
 - M-estimation (Huber, 1973),
 - S-estimation (Rousseeuw and Yohai, 1984), and
 - MM-estimation

2D and 3D Feature-based Alignment

Random Sample Consensus (RANSAC)

- ★ It is an iterative method to estimate parameters of a mathematical model from a set of observed data that contains outliers.
- ★ It also can be interpreted as an *outlier detection method*.
- ★ It is a non-deterministic algorithm in the sense that it produces a reasonable result only with a certain probability,
 - with this probability increasing as more iterations are allowed.

2D and 3D Feature-based Alignment

3D Alignment

- ★ Any computer vision applications require the alignment of 3D points.
- ★ Linear 3D transformations (e.g., for translation, similarity, and affine) can use regular least squares to estimate parameters.
- ★ Rigid motion:

$$E_{R3D} = \sum_i \|x_i' - R x_i - t\|^2$$

- ★ We can center the point clouds:

$$\{\hat{x}_i = x_i - c\}$$

$$\{\hat{x}_i' = x_i' - c'\}$$

- ★ Estimate the rotation between $\hat{x}_i$ and $\hat{x}_i'$

Pose Estimation

Introduction

- ★ A particular instance of feature-based alignment is estimating an object's 3D pose from a set of 2D point projections.
- ★ This pose estimation problem is also known as *extrinsic calibration*,
 - as opposed to the intrinsic calibration of internal camera parameters such as focal length.
- ★ Linear algorithms
- ★ Iterative algorithms

Pose Estimation

Camera Matrix

- ★ This matrix maps the 3-D world scene into the image plane.
- ★ The camera matrix is calculated with the use of the *extrinsic* and *intrinsic parameters*.
- ★ The *extrinsic parameters* represent the *location of the camera* in the 3-D scene.
- ★ The *intrinsic parameters* represent the *optical center and focal length* of the camera.

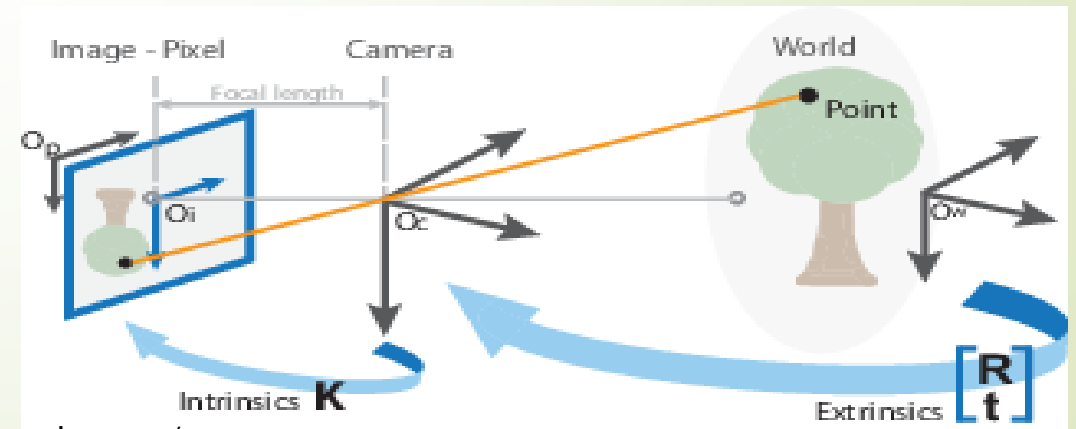
Pose Estimation

Camera Calibration Parameters

- ★ The calibration algorithm calculates the camera matrix using the extrinsic and intrinsic parameters.
- ★ The extrinsic parameters represent a **rigid transformation** from *3-D world coordinate system* to the *3-D camera's coordinate system*.
- ★ The intrinsic parameters represent a **projective transformation** from the *3-D camera's coordinates* into the *2-D image coordinates*.
- ★ The world points are transformed to camera coordinates using the extrinsics parameters.
- ★ The camera coordinates are mapped into the image plane using the intrinsics parameters.

$$P = \begin{bmatrix} R \\ t \end{bmatrix} K$$

Camera matrix *Extrinsics* *Intrinsic matrix*
Rotation and translation



Pose Estimation

Extrinsic parameters

- ★ The extrinsic parameters consist of a rotation, R , and a translation, t .
- ★ The origin of the camera's coordinate system is at its optical center and its x- and y-axis define the image plane.



Pose Estimation

Intrinsic parameters

- ★ The intrinsic parameters include the focal length, the optical center, also known as the principal point, and the skew coefficient.
- ★ The camera intrinsic matrix, K , is defined as:

$$\begin{bmatrix} f_x & 0 & 0 \\ s & f_y & 0 \\ c_x & c_y & 1 \end{bmatrix}$$

Pose Estimation

Linear Algorithms

- ★ Simplest way to recover the pose of the camera
- ★ Form a set of linear equations analogous to those used for 2D motion estimation from the camera matrix form of perspective projection.

$$\begin{aligned} x_i &= \frac{p_{00}X_i + p_{01}Y_i + p_{02}Z_i + p_{03}}{p_{20}X_i + p_{21}Y_i + p_{22}Z_i + p_{23}} \\ y_i &= \frac{p_{10}X_i + p_{11}Y_i + p_{12}Z_i + p_{13}}{p_{20}X_i + p_{21}Y_i + p_{22}Z_i + p_{23}} \end{aligned}$$

(x_i, y_i) : measured *2D feature locations*, (X_i, Y_i, Z_i) : known *3D feature locations*

- ★ Solve the camera matrix $\mathbf{P}$ in a linear fashion.
- ★ Multiply the denominator on both sides of the equation.
- ★ Denominator(D):

$$p_{20}X_i + p_{21}Y_i + p_{22}Z_i + p_{23}$$

Pose Estimation

Linear Algorithms

- ★ The resulting algorithm is called the *direct linear transform (DLT)*.
- ★ At least six correspondences are needed to compute the 12 (or 11) unknowns in $\mathbf{P}$.
- ★ More accurate estimation of $\mathbf{P}$ can be obtained by non-linear least squares with a small number of iterations.

$$\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$$

- ★ Recover both the intrinsic calibration matrix $\mathbf{K}$ and the rigid transformation $(\mathbf{R}, \mathbf{t})$.
- ★ $\mathbf{K}$ and $\mathbf{R}$ can be obtained from the front 3×3 sub-matrix of $\mathbf{P}$ using $\mathbf{RQ}$ factorization.

Pose Estimation

Iterative Algorithms

- ★ The most accurate (and flexible) way to estimate pose is using non-linear least squares
- ★ Projection equations:

$$\mathbf{x}_i = f(\mathbf{p}_i; \mathbf{R}, \mathbf{t}, \mathbf{K})$$

- ★ Iteratively minimize the robustified linearized reprojection errors:

$$E_{\text{NLP}} = \sum_i \rho \left(\frac{\partial f}{\partial \mathbf{R}} \Delta \mathbf{R} + \frac{\partial f}{\partial \mathbf{t}} \Delta \mathbf{t} + \frac{\partial f}{\partial \mathbf{K}} \Delta \mathbf{K} - \mathbf{r}_i \right)$$

Pose Estimation

Application: Augmented Reality

- ★ Human brain is doing pose estimation every single second.
- ★ Virtual 3D images or annotations are superimposed on top of a live video feed.
- ★ Either through the use of see through glasses (a head-mounted display) or on a regular computer or mobile device screen.



Geometric Intrinsic Calibration

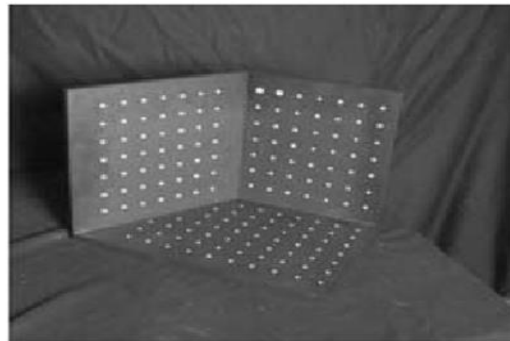
Introduction

- ★ The process of computing the camera parameters is called *camera calibration*.
- ★ Intrinsic and extrinsic calibration can be computed at the same time
 - Intrinsic calibration: *Internal camera calibration parameters*
 - Extrinsic calibration: *Pose of the camera*
- ★ The classic approach to camera calibration in both the photogrammetry and the computer vision communities.

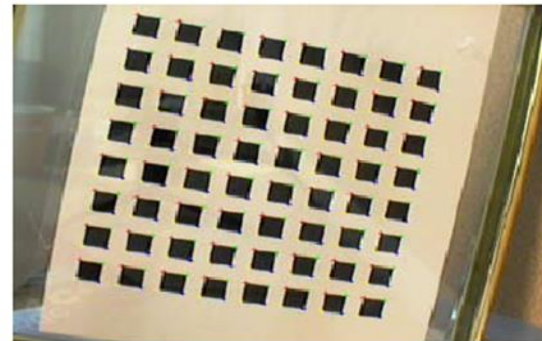
Geometric Intrinsic Calibration

Calibration Patterns

- ★ The use of a calibration pattern or set of markers is:
 - one of the more reliable ways to estimate a camera's intrinsic parameters.
- ★ In photogrammetry, it is common to set up a camera in a large field looking at distant calibration targets
 - whose exact location has been pre-computed using surveying equipment.



(a)



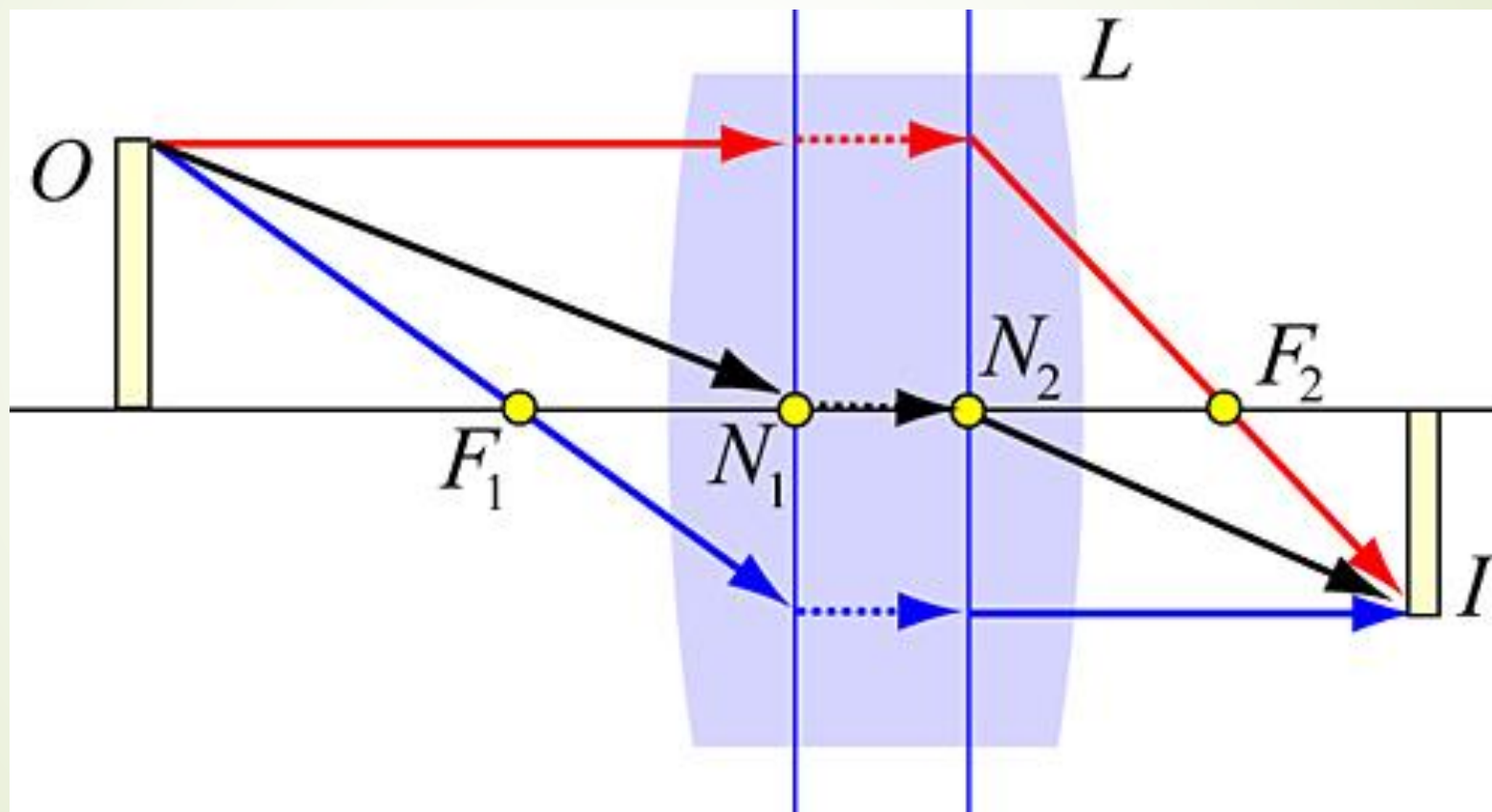
(b)

Figure 6.8 Calibration patterns: (a) a three-dimensional target (Quan and Lan 1999) © 1999 IEEE; (b) a two-dimensional target (Zhang 2000) © 2000 IEEE. Note that radial distortion needs to be removed from such images before the feature points can be used for calibration.

Geometric Intrinsic Calibration

Nodal Point

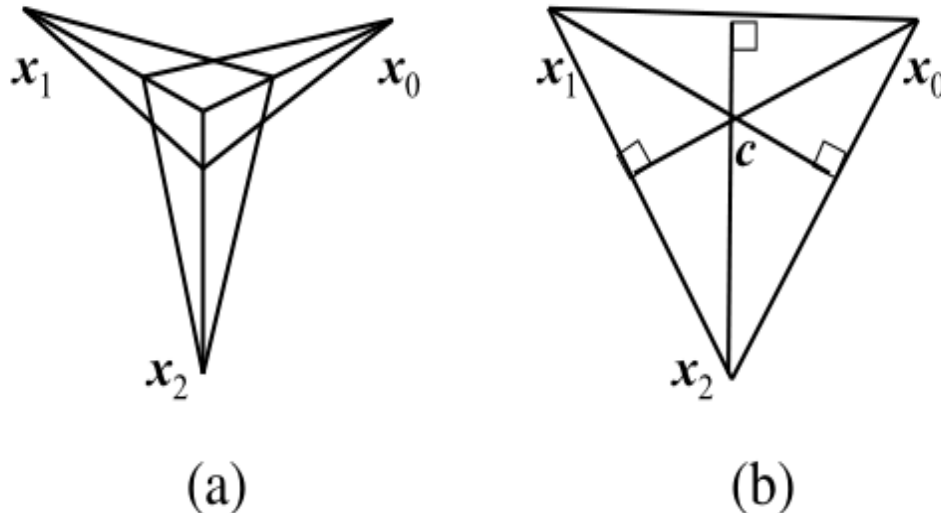
★ Nodal points N_1 and N_2 .



Geometric Intrinsic Calibration

Vanishing Points

- ★ A common case for calibration.
- ★ The camera is looking at a man-made scene with a lot of rectangular objects such as boxes or room walls.
- ★ Intersect the 2D lines corresponding to 3D parallel lines to compute their vanishing points.
- ★ Use these to determine the intrinsic and extrinsic calibration parameters.



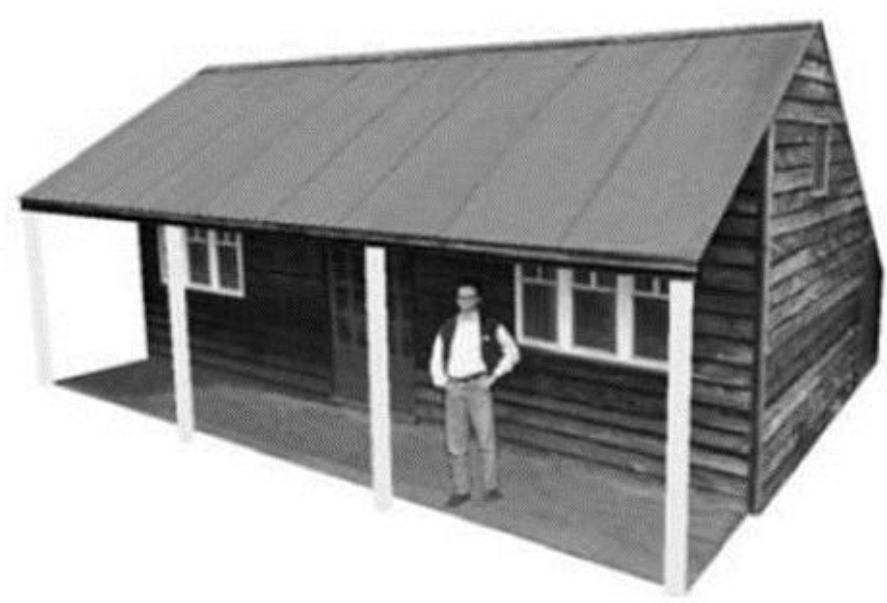
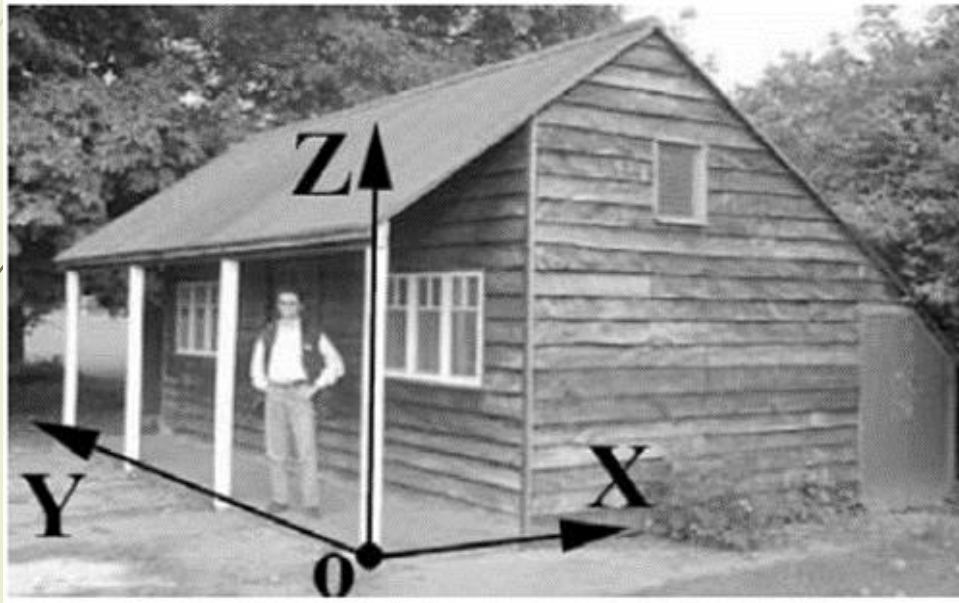
Calibration from vanishing points:

- (a) any pair of finite vanishing points can be used to estimate the focal length;
- (b) the orthocenter of the vanishing point triangle gives the optical center of the image c .

Geometric Intrinsic Calibration

Application: Single View Metrology

- ★ Allows people to interactively measure heights and other dimensions as well as to build piecewise planar 3D models.

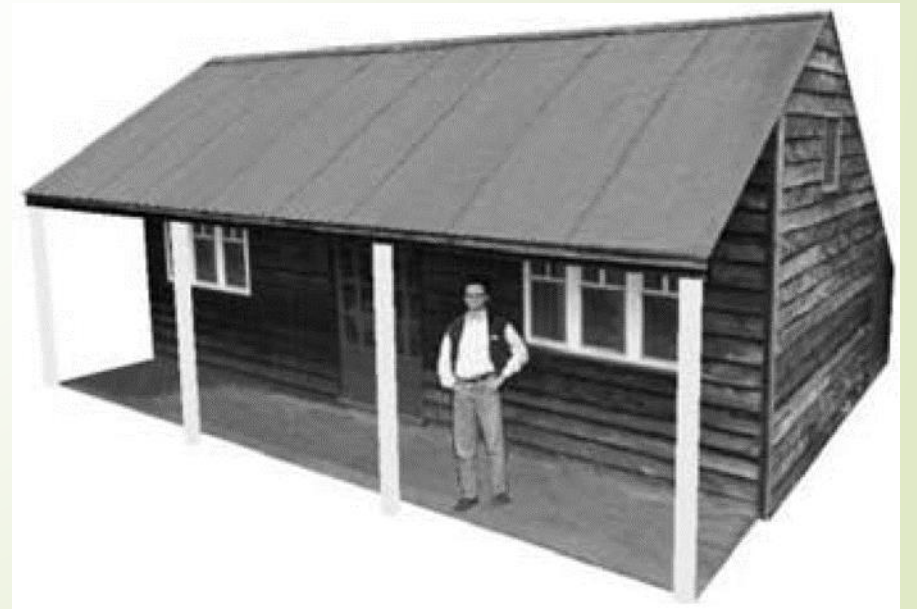
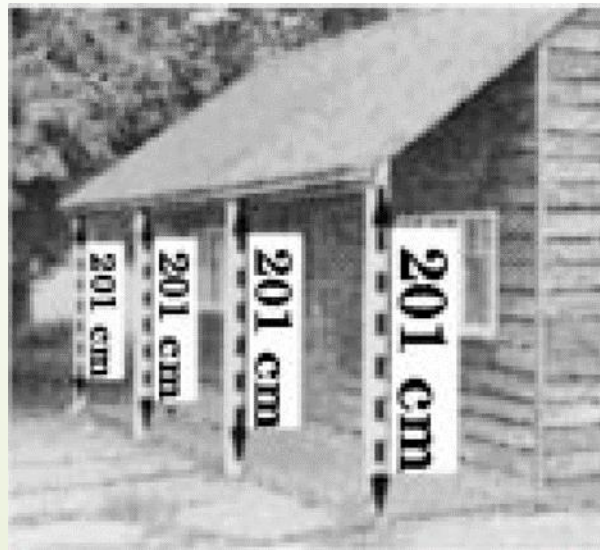
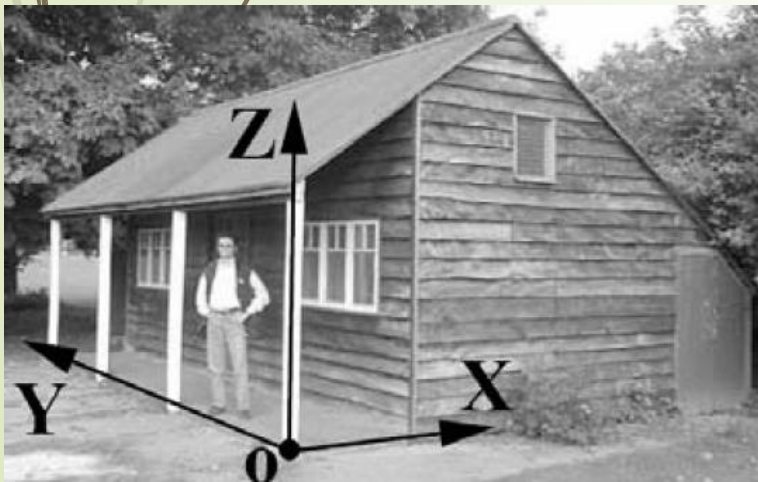


(a) input image showing the three coordinate axes computed from the two horizontal vanishing (b) a new view of the 3D reconstruction.

Geometric Intrinsic Calibration

Application: Single View Metrology

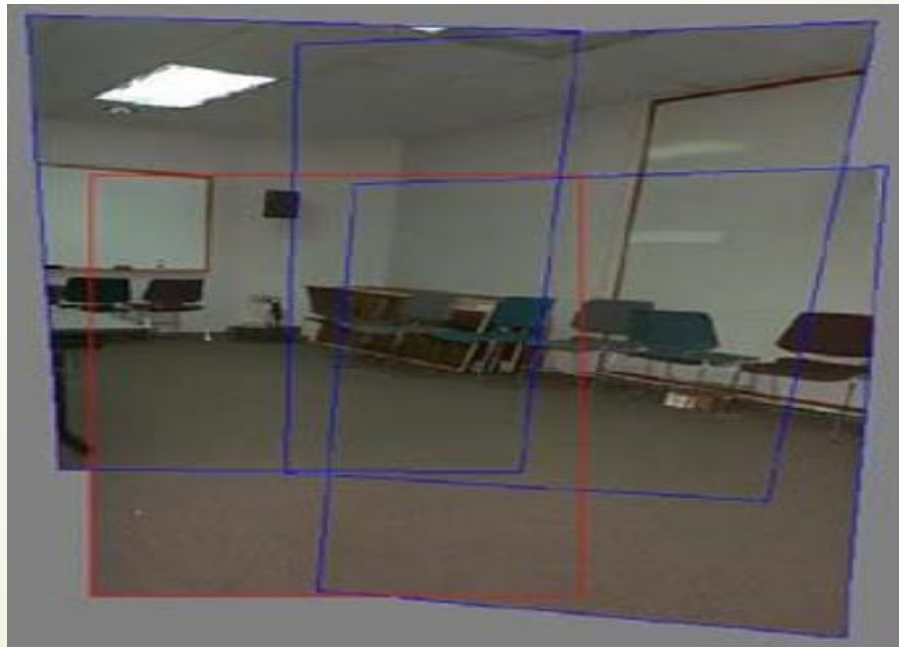
- ★ Criminisi, Reid, and Zisserman (2000).
- ★ First step: Identify two orthogonal vanishing points on the ground plane and the vanishing point for the vertical direction.
- ★ The user then marks a few dimensions in the image, such as the height of a reference object.
- ★ The system can automatically compute the height of another object. Walls and other planar impostors (geometry) can also be sketched.



Geometric Intrinsic Calibration

Rotational Motion

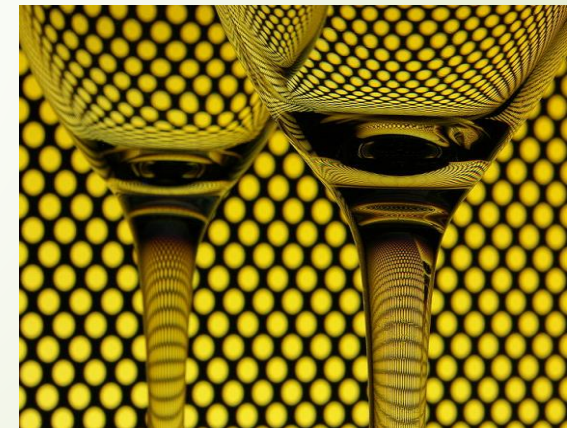
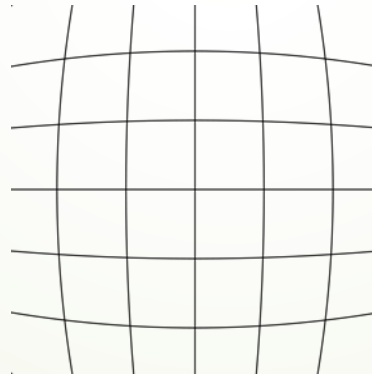
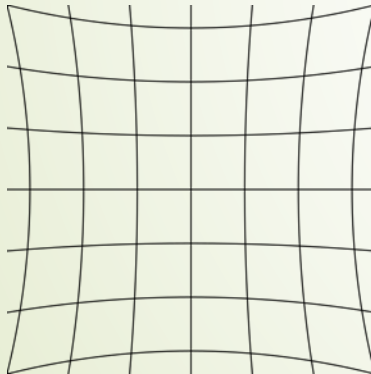
- ★ But you can rotate the camera around its front nodal point (or, equivalently, work in a large open environment where all objects are distant).
- ★ The camera can be calibrated from a set of overlapping images by assuming that it is undergoing pure rotational motion.
- ★ The accuracy in this estimate is proportional to the total number of pixels in the resulting cylindrical panorama.



Geometric Intrinsic Calibration

Radial Distortion

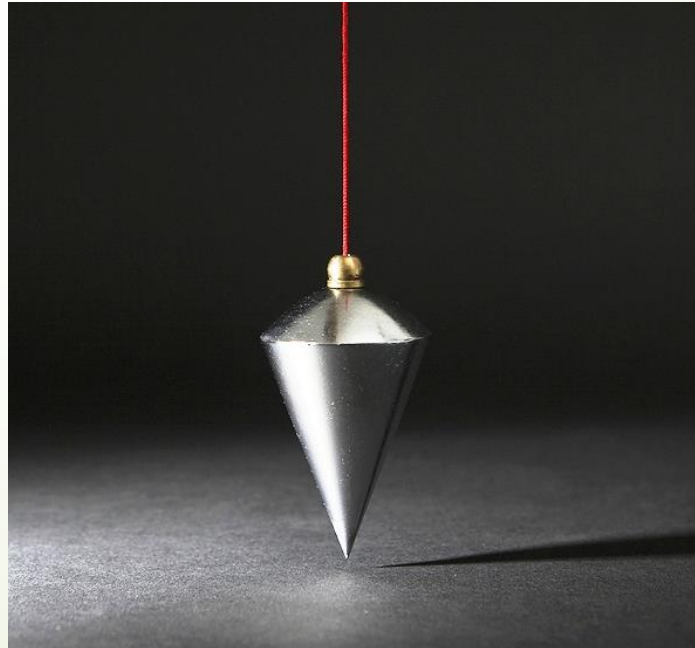
- ★ When images are taken with wide-angle lenses, it is often necessary to model lens distortions such as radial distortion.
 - barrel distortion
 - pincushion distortion



Geometric Intrinsic Calibration

Plumb Line Method

- ★ One of the simplest and most useful method.
- ★ Take an image of a scene with a lot of straight lines, especially lines aligned with near the edges of the image.
- ★ The radial distortion parameters can then be adjusted until all of the lines in the image are straight.



Geometric Intrinsic Calibration

Other Distortion

- ★ More general models of lens distortion, such as fisheye and non-central projection, may sometimes be required. While the parameterization of such lenses may be more complicated.
- ★ The general approach of either using
 - calibration rigs with known 3D positions or
 - self-calibration through the use of multiple overlapping images of a scene can both be used.

